Retail Sales Performance Analysis

A Statistical Business Intelligence Study Using Excel, MySQL, and Power BI

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Executive Summary

This report presents an end-to-end business intelligence analysis of the Sample Superstore dataset (9,994 orders, 2014–2017), covering data cleaning in Excel, relational querying in MySQL, statistical hypothesis testing, and dashboard development in Power BI. The analysis identifies a 15-point profitability gap between the Technology and Furniture categories despite comparable revenue, quantifies a statistically significant negative relationship between discount rate and profit (Pearson $r = -0.22$), and confirms that regional profit differences are statistically significant at the 5% level (one-way ANOVA, $p = 0.049$). These findings are translated into concrete business recommendations covering pricing, product mix, customer targeting, regional strategy, and logistics.

Contents

1 Project Overview

Retail organisations generate large volumes of transactional data containing valuable insight into customer purchasing behaviour, product performance, profitability, and operational efficiency. Systematic analysis of this data allows businesses to make informed decisions that increase revenue, control costs, and improve customer experience. This project analyses the Sample Superstore dataset using a three-tool workflow: Excel for data cleaning and exploratory analysis, MySQL for structured business querying, and Power BI for interactive dashboard reporting. Statistical methods — including descriptive statistics, Pearson correlation, and one-way ANOVA — are applied throughout to ensure that business conclusions are evidence-based rather than purely visual.

2 Business Problem Statement

A retail company wants to understand its sales performance, customer behaviour, product profitability, and regional operations more clearly. Management requires answers to specific business questions concerning revenue generation, profit optimisation, customer segmentation, discount effectiveness, and shipping efficiency, in order to support data-driven decision-making rather than relying on intuition. The company aims to use four years of historical transaction data (2014–2017) to identify recurring trends, isolate sources of profit leakage, and prioritise corrective action where it will have the greatest financial impact.

3 Objectives

- Analyse overall sales and profitability across the full dataset.
- Evaluate category and sub-category performance, identifying high-revenue but low-margin product lines.
- Compare regional sales and profit performance across all four regions.
- Examine customer segment contributions to revenue and profit.
- Analyse the relationship between discount rate and profitability.
- Study monthly and yearly sales trends to identify seasonality.
- Evaluate shipping mode efficiency and delivery performance.
- Apply statistical methods (correlation, ANOVA) to validate that observed business patterns are statistically significant rather than incidental.
- Develop a three-page interactive Power BI dashboard to support ongoing decision-making.

4 Dataset Summary

Dataset: Sample Superstore Dataset
Period covered: January 2014 – December 2017

Metric	Value
Records	9,994
Original Columns	15
Engineered Columns Added	6
Categories	3 (Technology, Furniture, Office Supplies)
Sub-categories	17
Regions	4 (East, West, Central, South)
Customer Segments	3 (Consumer, Corporate, Home Office)
Tools Used	Excel, MySQL, Power BI

Table 1: Dataset Overview

Scope and Limitations

This analysis is confined to the Sample Superstore transactional dataset covering 2014–2017 and does not incorporate external market data, marketing spend, or competitor pricing. Findings reflect historical patterns within this dataset and should be validated against current operational data before being used for live pricing or inventory decisions.

5 Data Preparation and Analysis in Excel

Excel was used to clean, transform, and perform exploratory analysis on the dataset before it was loaded into MySQL for relational querying.

5.1 Data Cleaning

- Verified completeness — confirmed no missing values across all 9,994 records.
- Converted `Order Date` and `Ship Date` from text to proper date format (DD-MM-YYYY to a true Excel date serial) to enable chronological sorting and date-based formulas.
- Removed duplicate records using `Remove Duplicates`.
- Standardised currency and percentage formatting across Sales, Profit, and Discount columns.

5.2 Feature Engineering

Six calculated columns were added to support downstream analysis:

- **Profit Margin** — Profit divided by Sales, expressed as a percentage.
- **Days to Ship** — difference between Ship Date and Order Date.
- **Year** and **Month Number** — extracted using `YEAR()` and `MONTH()` for time-based aggregation.
- **Month Name** — generated using `TEXT()` for readable chart labelling.
- **Discount Band** — a nested `IF()` classification grouping orders into six bands (No Discount, 1–10%, 11–20%, 21–30%, 31–50%, 51–80%).

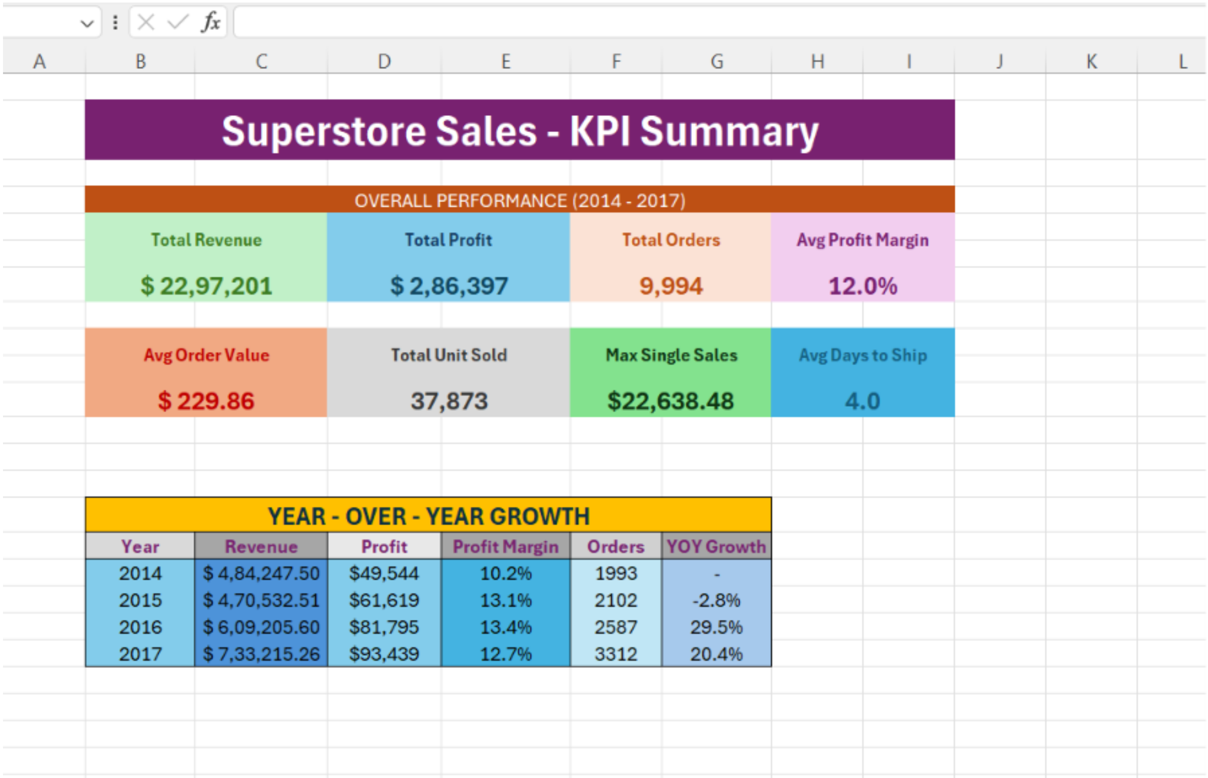
P	Q	R	S	T	U	V
Prifit Margin	Days to Ship	Year	Month Numbe	Month Name	Discount Band	
26%	6	2014	1	Jan	0% (No Discount)	
29%	6	2014	1	Jan	0% (No Discount)	
30%	6	2014	1	Jan	0% (No Discount)	
27%	6	2014	1	Jan	0% (No Discount)	
47%	6	2014	1	Jan	0% (No Discount)	
21%	6	2014	1	Jan	0% (No Discount)	
48%	5	2014	1	Jan	0% (No Discount)	
30%	5	2014	1	Jan	0% (No Discount)	
29%	6	2014	1	Jan	0% (No Discount)	
11%	6	2014	1	Jan	0% (No Discount)	
47%	6	2014	1	Jan	0% (No Discount)	
38%	6	2014	1	Jan	1-10%	
24%	6	2014	1	Jan	0% (No Discount)	
29%	6	2014	1	Jan	0% (No Discount)	
33%	6	2014	1	Jan	0% (No Discount)	
45%	6	2014	1	Jan	0% (No Discount)	
-77%	6	2014	1	Jan	51-80%	
16%	2	2014	1	Jan	11-20%	
48%	5	2014	1	Jan	0% (No Discount)	
0%	4	2014	1	Jan	0% (No Discount)	
49%	5	2014	1	Jan	0% (No Discount)	
0%	5	2014	1	Jan	0% (No Discount)	
48%	5	2014	1	Jan	0% (No Discount)	
50%	5	2014	1	Jan	0% (No Discount)	
0%	5	2014	1	Jan	0% (No Discount)	
36%	5	2014	1	Jan	0% (No Discount)	
28%	5	2014	1	Jan	0% (No Discount)	
47%	3	2014	1	Jan	0% (No Discount)	
28%	3	2014	1	Jan	0% (No Discount)	
25%	6	2014	1	Jan	0% (No Discount)	

Figure 1: Additional Columns Created in Excel

5.3 KPI Analysis

The following top-line metrics were calculated to give management an at-a-glance view of business performance:

- Total Revenue — $2,297,200
- Total Profit — $286,397
- Total Orders — 9,994
- Total Quantity Sold
- Average Order Value — $229.86
- Average Profit Margin — 12.5%
- Maximum Single Sale
- Average Days to Ship



Superstore Sales - KPI Summary

OVERALL PERFORMANCE (2014 - 2017)

Total Revenue	Total Profit	Total Orders	Avg Profit Margin
$ 22,97,201	$ 2,86,397	9,994	12.0%
Avg Order Value	Total Unit Sold	Max Single Sales	Avg Days to Ship
$ 229.86	37,873	$22,638.48	4.0

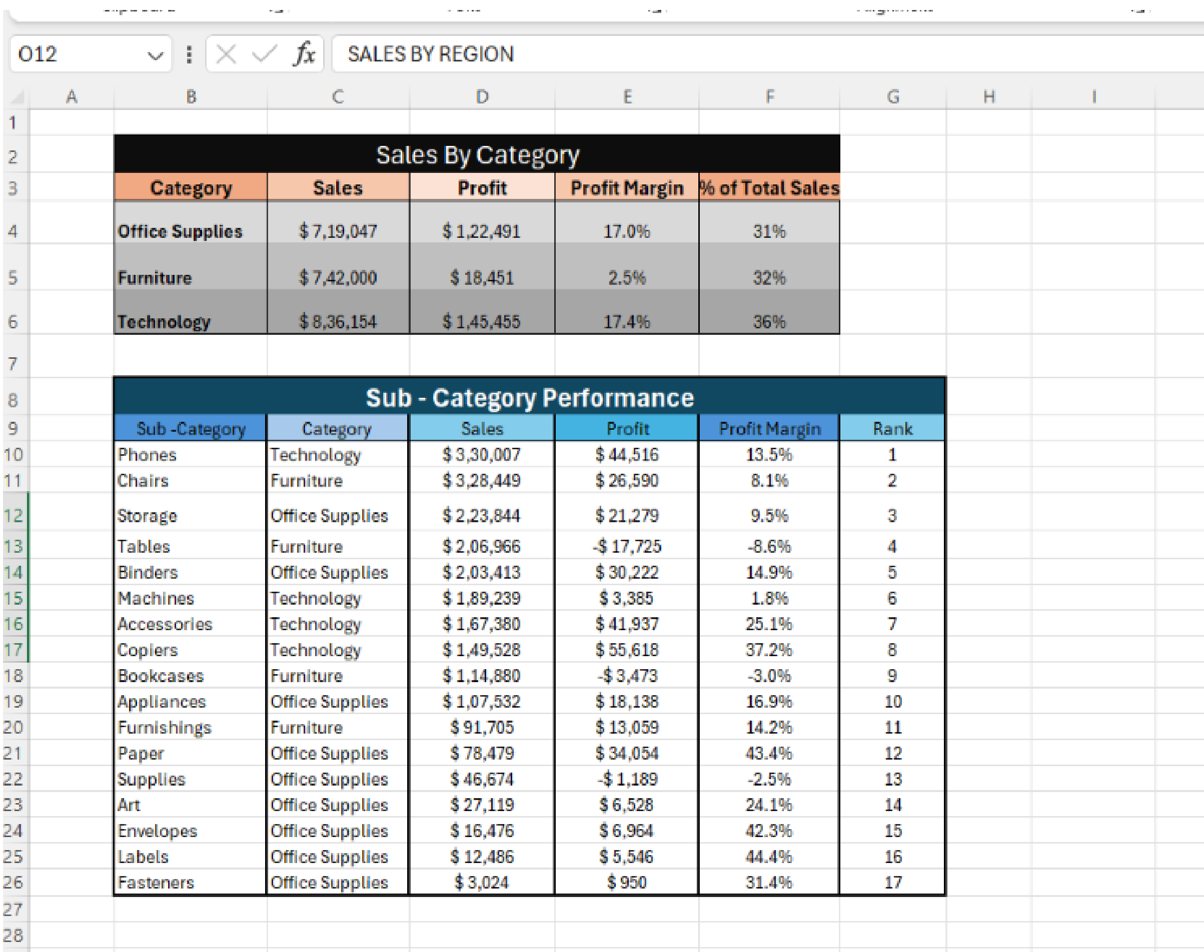
YEAR - OVER - YEAR GROWTH

Year	Revenue	Profit	Profit Margin	Orders	YOY Growth
2014	$ 4,84,247.50	$49,544	10.2%	1993	-
2015	$ 4,70,532.51	$61,619	13.1%	2102	-2.8%
2016	$ 6,09,205.60	$81,795	13.4%	2587	29.5%
2017	$ 7,33,215.26	$93,439	12.7%	3312	20.4%

Figure 2: Excel KPI Dashboard

5.4 Category Performance Analysis

Technology generated the highest revenue ($836,154) with a healthy 17.4% profit margin. Furniture generated comparable revenue ($741,999) but only a 2.5% margin — a 15-point gap driven primarily by the Tables and Bookcases sub-categories, both of which operate at a net loss. The figure below shows sales, profit, and quantity sold broken down by category and ranked sub-category.



O12 SALES BY REGION

Sales By Category

Category	Sales	Profit	Profit Margin	% of Total Sales
Office Supplies	$ 7,19,047	$ 1,22,491	17.0%	31%
Furniture	$ 7,42,000	$ 18,451	2.5%	32%
Technology	$ 8,36,154	$ 1,45,455	17.4%	36%

Sub - Category Performance

Sub -Category	Category	Sales	Profit	Profit Margin	Rank
Phones	Technology	$ 3,30,007	$ 44,516	13.5%	1
Chairs	Furniture	$ 3,28,449	$ 26,590	8.1%	2
Storage	Office Supplies	$ 2,23,844	$ 21,279	9.5%	3
Tables	Furniture	$ 2,06,966	-$ 17,725	-8.6%	4
Binders	Office Supplies	$ 2,03,413	$ 30,222	14.9%	5
Machines	Technology	$ 1,89,239	$ 3,385	1.8%	6
Accessories	Technology	$ 1,67,380	$ 41,937	25.1%	7
Copiers	Technology	$ 1,49,528	$ 55,618	37.2%	8
Bookcases	Furniture	$ 1,14,880	-$ 3,473	-3.0%	9
Appliances	Office Supplies	$ 1,07,532	$ 18,138	16.9%	10
Furnishings	Furniture	$ 91,705	$ 13,059	14.2%	11
Paper	Office Supplies	$ 78,479	$ 34,054	43.4%	12
Supplies	Office Supplies	$ 46,674	-$ 1,189	-2.5%	13
Art	Office Supplies	$ 27,119	$ 6,528	24.1%	14
Envelopes	Office Supplies	$ 16,476	$ 6,964	42.3%	15
Labels	Office Supplies	$ 12,486	$ 5,546	44.4%	16
Fasteners	Office Supplies	$ 3,024	$ 950	31.4%	17

Figure 3: Category Sales and Profit Analysis

5.5 Regional Analysis

The West region generated the highest revenue and profit margin of all four regions, while the Central region consistently underperformed across both metrics. The figure below presents sales, profit, and order count for each of the four regions.

SALES BY REGION

Region	Sales	Profit	Profit Margin	Orders
Central	$ 5,01,240	$ 39,706	7.9%	2323
East	$ 6,78,781	$ 91,523	13.5%	2848
West	$ 7,25,458	$ 1,08,418	14.9%	3203
South	$ 3,91,722	$ 46,749	11.9%	1620

Figure 4: Regional Performance Analysis

5.6 Customer Segment Analysis

The Consumer segment contributed the largest share of revenue (51%), followed by Corporate (31%) and Home Office (18%). The segment-by-category crosstab below shows that Consumer customers concentrate spend in Technology, while Corporate and Home Office customers favour Office Supplies.

Customer Segment Analysis

Segment	Sales	% Revenue	Profit	Profit Margin	Orders	Avg Order Value
Consumer	$11,61,401	50.6%	$1,34,119	11.5%	5191	$224
Home Office	$4,29,653	18.7%	$60,299	14.0%	1783	$241
Corporate	$7,06,146	30.7%	$91,979	13.0%	3020	$234

SEGMENT x CATEGORY CROSSTAB (Sales)

Segment	Furniture	Office Supplies	Technology	Total
Corporate	$2,29,020	$2,30,676	$2,46,450	$7,06,146
Home Office	$1,21,931	$1,24,418	$1,83,304	$4,29,653
Consumer	$3,91,049	$3,63,952	$4,06,400	$11,61,401

SEGMENT X CATEGORY

Row Labels	Sum of Sales
Consumer	**$11,61,401.35**
Furniture	$3,91,049.31
Office Supplies	$3,63,952.14
Technology	$4,06,399.90
Corporate	**$7,06,146.37**
Furniture	$2,29,019.79
Office Supplies	$2,30,676.46
Technology	$2,46,450.12
Home Office	**$4,29,653.15**
Furniture	$1,21,930.70
Office Supplies	$1,24,418.43
Technology	$1,83,304.02
Grand Total	**$22,97,200.86**

Figure 5: Customer Segment Performance and Segment × Category Crosstab

5.7 Monthly Trend Analysis

Monthly aggregation across all four years reveals a consistent seasonal pattern, with revenue peaking in the fourth quarter (October–December) of every year and reaching its lowest point in February. The figure below shows the 48-month trend line with a colour scale highlighting peak and trough months.

Monthly Sales Trend (2014 - 2017)						
Year	Month	Month Name	Sales	Profit	Quantity	Profit Margin
2014	1	Jan	$ 14,237	$ 2,450	284	17.2%
2014	2	Feb	$ 4,520	$ 862	159	19.1%
2014	3	Mar	$ 55,691	$ 499	585	0.9%
2014	4	Apr	$ 28,295	$ 3,489	536	12.3%
2014	5	May	$ 23,648	$ 2,739	466	11.6%
2014	6	Jun	$ 34,595	$ 4,977	521	14.4%
2014	7	Jul	$ 33,946	-$ 841	550	-2.5%
2014	8	Aug	$ 27,909	$ 5,318	609	19.1%
2014	9	Sep	$ 81,777	$ 8,328	1000	10.2%
2014	10	Oct	$ 31,453	$ 3,448	573	11.0%
2014	11	Nov	$ 78,629	$ 9,292	1219	11.8%
2014	12	Dec	$ 69,546	$ 8,984	1079	12.9%
2015	1	Jan	$ 18,174	-$ 3,281	236	-18.1%
2015	2	Feb	$ 11,951	$ 2,814	239	23.5%
2015	3	Mar	$ 38,726	$ 9,732	515	25.1%
2015	4	Apr	$ 34,195	$ 4,187	543	12.2%
2015	5	May	$ 30,132	$ 4,668	575	15.5%
2015	6	Jun	$ 24,797	$ 3,336	486	13.5%
2015	7	Jul	$ 28,765	$ 3,289	557	11.4%
2015	8	Aug	$ 36,898	$ 5,356	598	14.5%
2015	9	Sep	$ 64,596	$ 8,209	1086	12.7%
2015	10	Oct	$ 31,405	$ 2,817	631	9.0%
2015	11	Nov	$ 75,973	$ 12,475	1310	16.4%
2015	12	Dec	$ 74,920	$ 8,017	1203	10.7%
2016	1	Jan	$ 18,542	$ 2,825	358	15.2%
2016	2	Feb	$ 22,979	$ 5,005	306	21.8%
2016	3	Mar	$ 51,716	$ 3,612	579	7.0%
2016	4	Apr	$ 38,750	$ 2,978	635	7.7%
2016	5	May	$ 56,988	$ 8,662	863	15.2%
2016	6	Jun	$ 40,345	$ 4,750	742	11.8%
2016	7	Jul	$ 39,262	$ 4,433	758	11.3%
2016	8	Aug	$ 31,115	$ 2,062	693	6.6%

Figure 6: Monthly Sales and Profit Trends, 2014–2017

5.8 Discount Impact Analysis

Orders with no discount or discounts up to 20% remain profitable, with the 1–10% band showing the highest average profit per order ($96.06). Beyond a 30% discount threshold, average profit per order turns negative, falling to as low as −$156.28 in the 31–50% band. The figure below shows average sales, profit, and margin for each discount band, with negative-margin bands highlighted.

DISCOUNT & PROFIT IMPACT ANALYSIS

Discount Band	Orders	Avg Sales	Avg Profit	Avg Margin	% Orders
0% (No Discount)	4798	$ 226.74	$ 66.90	34.0%	48.0%
1-10%	94	$ 578.40	$ 96.06	15.6%	0.9%
11-20%	3709	$ 213.58	$ 24.74	17.5%	37.1%
21-30%	227	$ 454.74	-$ 45.68	-11.5%	2.3%
31-50%	310	$ 630.05	-$ 156.28	-29.6%	3.1%
51-80%	856	$ 75.03	-$ 89.44	-113.9%	8.6%

INSIGHTS: High discounts (>30 %) destroy profit margins

Figure 7: Discount Band Analysis

5.9 Statistical Analysis

Descriptive statistics were computed for Sales and Profit using Excel's Data Analysis ToolPak, including mean, median, standard deviation, variance, range, skewness, and kurtosis. Sales data exhibits strong positive skewness, with the mean ($229.86) considerably higher than the median ($54.49) — indicating that a small number of large orders disproportionately drive total revenue.

DeSCRIPTIVE STATISTICS

Sales		*Profit*	
Mean	229.8580008	Mean	28.65689631
Standard Error	6.234321582	Standard Error	2.343304174
Median	54.49	Median	8.6665
Mode	12.96	Mode	0
Standard Deviation	623.2451005	Standard Deviation	234.2601077
Sample Variance	388434.4553	Sample Variance	54877.79806
Kurtosis	305.3117532	Kurtosis	397.1885146
Skewness	12.97275234	Skewness	7.561431562
Range	22638.036	Range	14999.954
Minimum	0.444	Minimum	-6599.978
Maximum	22638.48	Maximum	8399.976
Sum	2297200.86	Sum	286397.0217
Count	9994	Count	9994

CORRELATION ANALYSIS

CORRELATION= -0.219487456

Statistical Inference: Negetive correlation between discount and profit

Interpretation: As discount increases, profit decreases

Figure 8: Descriptive Statistics for Sales and Profit

A Pearson correlation between Discount and Profit was also calculated, yielding $r = -0.22$, confirming a statistically meaningful negative relationship between the two variables.

5.10 ANOVA Analysis

A one-way ANOVA was conducted to test whether mean profit differs significantly across the four regions.

H_0: Mean profit is equal across all four regions.
H_1: Mean profit differs across at least one region.

The test returned $F = 2.62$ and $p = 0.049$, which is below the 5% significance threshold. The null hypothesis is therefore rejected: regional profit differences are statistically significant, with the West region showing the highest average profit per order and the Central region the lowest — a difference of approximately $16.76 per order. The figure below shows the full ANOVA summary and source-of-variation table.

ANOVA REGIONAL ANALYSIS

East	West	Central	South
4.884	9.3312	5.5512	5.2398
3.0814	238.653	4.2717	746.4078
13.0928	3.9294	-64.7748	274.491
-2.5212	6.567	-5.487	1.4796
-53.2856	6.4864	-53.7096	113.6742
28.5984	-23.716	-18.2525	204.1092
-31.05	19.6224	1.168	0.3112
-13.6312	36.693	9.75	3.0084
10.1493	-320.597	4.0542	1.3583
1.3257	9.2928	30.0234	21.2954
49.6048	5.79	8.9535	5.3392
9.744	19.2384	93.5816	25.47
3.4048	14.3075	15.9792	258.696
1.6038	-16.65	12.7368	2.7072
5.3392	12.0252	4.8609	34.3548
17.472	3.4196	13.365	87.3504
1.07	5.8045	0	65.978
32.13	27.248	5.0196	181.9818
4.5954	4.3134	39.7488	6.6584
40.872	29.0136	4.752	11.58
9.1785	16.0928	6.8982	1.7901
4.914	6.0368	206.316	10.9698
-199.617	5.5328	-14.7708	224.2674
-10.0512	-24.294	15.2225	-51.5154
15.4752	86.3892	6.8724	4.1028
177.5889	22.9488	4.4344	30.7818
5.3998	4.7724	53.2704	0
16.7508	7.209	24.393	9.3312
45.9754	6.2208	2.5984	8.34
2.288	9.3312	6.066	0
1.7004	11.1024	-1.728	4.4712

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
East	2848	91522.78	32.13580758	63305.11842
West	3203	108418.4489	33.84903181	30313.97192
Central	2323	39706.3625	17.09270878	84964.94207
South	1620	46749.4303	28.85767302	45323.46695

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	431537.28	3	143845.7616	2.622478155	0.048892	2.605798
Within Groups	547962299	9990	54851.08095			
Total	548393836	9993				

H0: Mean profit is equal across all 4 regions
H1: Mean profit differs across regions

Test: One-Way ANOVA
Significance Level: α = 0.05
F-statistic: 2.62247
P-value: 0.04889
F critical: 2.6058

CONCLUSION:
Since **p-value (0.04889)** < **α (0.05)** and **F (2.62247)** > **F critical (2.6058)**,
we **REJECT** the null hypothesis(**H0**).

Figure 9: One-Way ANOVA Results — Profit by Region

6 Data Analysis Using MySQL

The cleaned dataset (9,994 rows) was loaded into a MySQL database (`superstore_db`) to support structured, repeatable business querying beyond what spreadsheet formulas can comfortably handle. Over twenty queries were written across five analytical areas: KPI summary, category and regional performance, monthly and quarterly trends, customer segment analysis, and discount impact. Each subsection below presents the query logic alongside the corresponding result set.

6.1 KPI Summary Queries

Aggregate business KPIs — total revenue, total profit, profit margin, and order count — were calculated using `SUM`, `AVG`, and `ROUND`. Year-over-year revenue and growth percentage were calculated using the `LAG()` window function to compare each year against the previous one.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_revenue	total_Profit	total_orders	total_units	avg_order_value	profit_margin_pct
2297201.07	286397.79	9994	37873	229.86	12.5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	revenue	growth_pct
2014	484247.56	NULL
2015	470532.46	-2.8
2016	609205.86	29.5
2017	733215.19	20.4

Figure 10: SQL Output — Overall KPIs and Year-over-Year Growth

6.2 Category and Regional Analysis Queries

Category and sub-category performance were calculated using `GROUP BY` with a window-function rank (`RANK() OVER (ORDER BY SUM(sales) DESC)`) to order sub-categories by revenue. Loss-making sub-categories were isolated using `GROUP BY` combined with `HAVING SUM(profit) < 0`, and regional performance was calculated using the same aggregation pattern applied to the `region` column.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Category	revenue	profit	margin_pct	pct_of_total
Technology	836154.10	145455.66	17.4	36.4
Furniture	741999.98	18451.25	2.5	32.3
Office Supplies	719046.99	122490.88	17.0	31.3

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Sub_Category	Category	revenue	profit	revenue_rank
Phones	Technology	330007.10	44516.25	1
Chairs	Furniture	328449.13	26590.15	2
Storage	Office Supplies	223843.59	21279.05	3
Tables	Furniture	206965.68	-17725.59	4
Binders	Office Supplies	203412.77	30221.64	5
Machines	Technology	189238.68	3384.73	6
Accessories	Technology	167380.31	41936.78	7
Copiers	Technology	149528.01	55617.90	8
Bookcases	Furniture	114880.05	-3472.56	9
Appliances	Office Supplies	107532.14	18138.07	10
Furnishings	Furniture	91705.12	13059.25	11

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Region	revenue	profit	profit_margin	Orders
Central	501239.88	39706.45	7.9	2323
East	678781.36	91522.84	13.5	2848
West	725457.93	108418.79	14.9	3203
South	391721.90	46749.71	11.9	1620

Figure 11: SQL Output — Category, Sub-Category Ranking, and Regional Performance

6.3 Monthly and Quarterly Trend Queries

Monthly revenue and profit were calculated by grouping on `YEAR(order_date)` and `MONTH(order_date)`, with `MONTHNAME()` used for readable labelling. A quarterly summary was produced using a `CASE WHEN` statement to map each month to its corresponding quarter, and the single best and worst performing months were identified using a correlated subquery combined with `ORDER BY` and `LIMIT 1`.

year	month_no	month_name	revenue	profit	orders
2014	1	January	14236.90	2450.18	79
2014	2	February	4519.92	862.30	46
2014	3	March	55691.04	498.72	157
2014	4	April	28295.35	3488.86	135
2014	5	May	23648.28	2738.74	122
2014	6	June	34595.14	4976.56	135
2014	7	July	33946.37	-841.46	143
2014	8	August	27909.47	5318.11	153
2014	9	September	81777.34	8328.08	268
2014	10	October	31453.37	3448.23	159

year	quarter	revenue	profit	orders
2014	Q1	74447.86	3811.20	282
2014	Q2	86538.77	11204.16	392
2014	Q3	143633.18	12804.73	564
2014	Q4	179627.75	21723.97	755
2015	Q1	68851.74	9264.94	260
2015	Q2	89124.28	12190.92	444
2015	Q3	130259.51	16853.70	592
2015	Q4	182296.93	23309.13	806
2016	Q1	93237.20	11441.39	335
2016	Q2	136082.33	16390.30	594

label	year	month_name	revenue
best month	2017	November	118447.81
worst month	2014	February	4519.92

Figure 12: SQL Output — Monthly Trend, Quarterly Summary, and Best/Worst Month

6.4 Customer Segment Queries

Segment-level revenue, profit, and average order value were calculated using `GROUP BY segment`. A Segment × Category crosstab was then built using conditional aggregation (`SUM(CASE WHEN category = '...' THEN sales ELSE 0 END)`), pivoting the three category totals into separate columns for each segment so that spend patterns across Consumer, Corporate, and Home Office customers could be compared side by side.

Segment	orders	revenue	profit	margin_pct
Consumer	5191	1161401.34	134119.33	11.5
Corporate	3020	706146.44	91979.45	13.0
Home Office	1783	429653.29	60299.01	14.0

Segment	technology	furniture	office_suppies	total
Consumer	406399.95	391049.43	363951.96	1161401.34
Corporate	246450.12	229019.83	230676.49	706146.44
Home Office	183304.03	121930.72	124418.54	429653.29

Figure 13: SQL Output — Segment Performance and Segment × Category Crosstab

6.5 Discount Impact Queries

Orders were grouped into discount bands using a `CASE WHEN` classification replicating the Excel `Discount Band` logic, and average sales, average profit, and average margin were calculated for each band using `AVERAGEIF`-equivalent `GROUP BY` aggregation. A supporting query using `WHERE profit < 0` was used to isolate and inspect individual loss-making orders directly.

discount_band	orders	revenue	profit	margin_pct
No Discound	4798	1087908.47	320987.88	29.5
1-10%	94	54369.30	9029.21	16.6
11-20%	3709	792152.87	91757.14	11.6
21-30%	227	103226.76	-10369.34	-10.0
31-50%	310	195314.91	-48447.87	-24.8
51-80%	856	64228.76	-76559.23	-119.2

discount_level	avg_qty_sold	avg_profit	orders
No Discount	3.8	66.90	4798
Low (1-20%)	3.7	26.50	3803
Medium (21-40%)	3.8	-77.86	460
High(41%+)	3.9	-106.71	933

Product_name	Category	discount	region	sales	profit
Cubify CubeX 3D Printer Double Head Print	Technology	0.70	East	4499.99	-6599.98
Cubify CubeX 3D Printer Triple Head Print	Technology	0.50	South	7999.98	-3839.99
GBC DocuBind P400 Electric Binding System	Office Supplies	0.80	Central	2177.58	-3701.89
Lexmark MX611dhe Monochrome Laser Printer	Technology	0.70	West	2549.99	-3399.98
Ibico EPK-21 Electric Binding System	Office Supplies	0.80	Central	1889.99	-2929.48
Cubify CubeX 3D Printer Double Head Print	Technology	0.70	East	1799.99	-2639.99
Fellowes PB500 Electric Punch Plastic Comb Bind...	Office Supplies	0.80	Central	1525.19	-2287.78
Chromcraft Bull-Nose Wood Oval Conference T...	Furniture	0.40	South	4297.64	-1862.31
GBC DocuBind P400 Electric Binding System	Office Supplies	0.80	Central	1088.79	-1850.95
Cisco TelePresence System EX90 Videoconferen...	Technology	0.50	South	22638.48	-1811.08

Figure 14: SQL Output — Discount Band Analysis and Loss-Making Orders

7 Power BI Dashboard Development

A three-page interactive dashboard was built in Power BI, connected to the cleaned dataset, with DAX measures for revenue, profit, margin, order count, and average order value. Slicers for Year, Region, and Category allow drill-down filtering across every visual on each page.

7.1 Dashboard Page 1 – Executive Overview

KPI cards for Total Revenue, Total Profit, Profit Margin, Total Orders, and Average Order Value, alongside a yearly revenue trend line and a category-level bar chart.

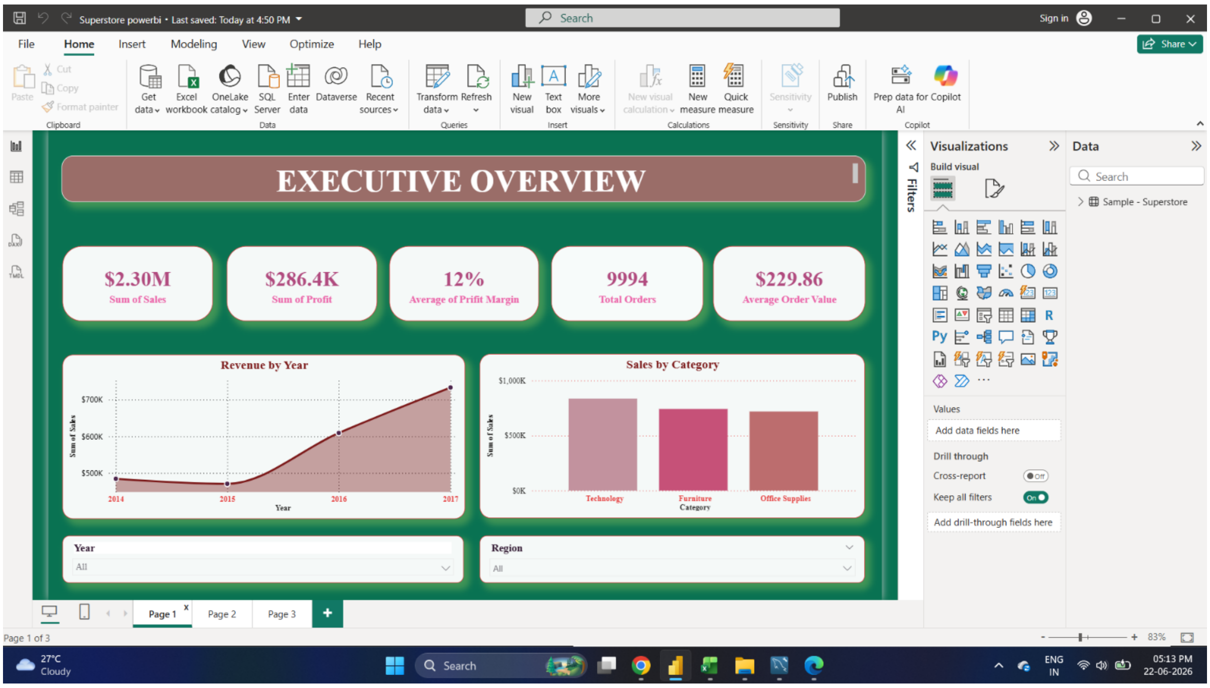


Figure 15: Executive Dashboard

7.2 Dashboard Page 2 – Customer and Regional Analysis

A geographic breakdown of sales by state, a Category $\times$ Region matrix, a segment donut chart, and a sub-category ranking visual.

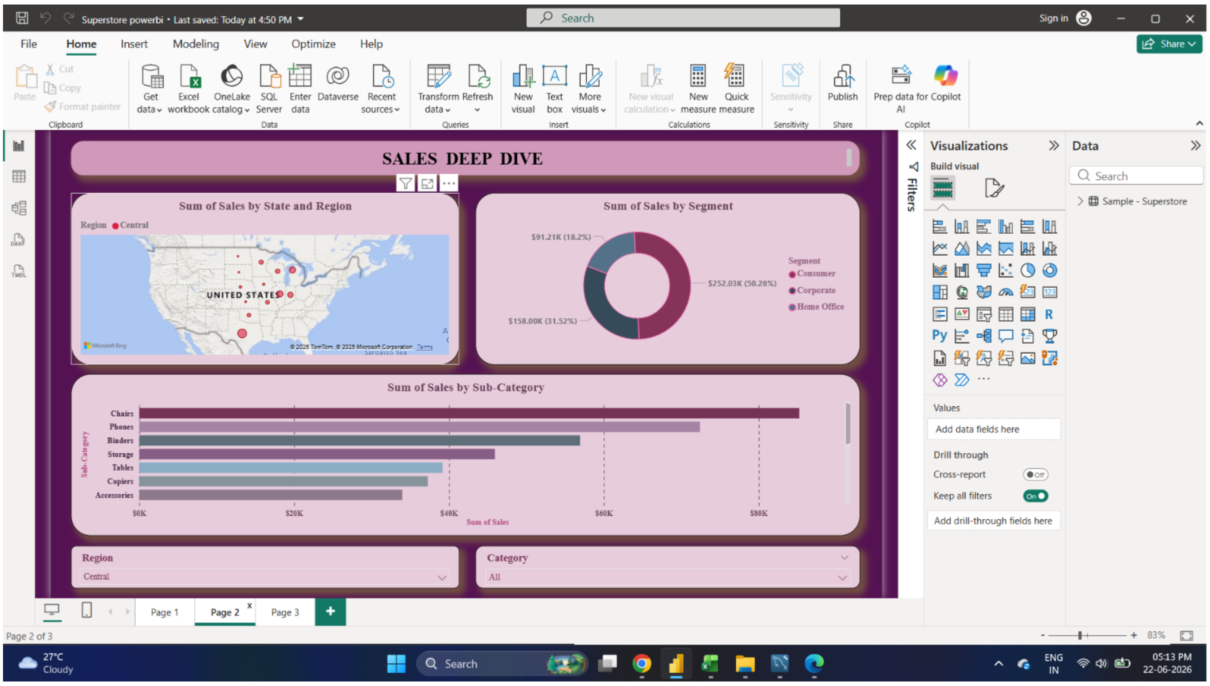


Figure 16: Customer and Regional Analysis Dashboard

7.3 Dashboard Page 3 – Product and Discount Analysis

A monthly profit trend line, an average profit by discount band chart with conditional red/green colour formatting, and a written summary of the statistical findings (correlation and ANOVA results) for decision-makers who do not work directly with the underlying data.

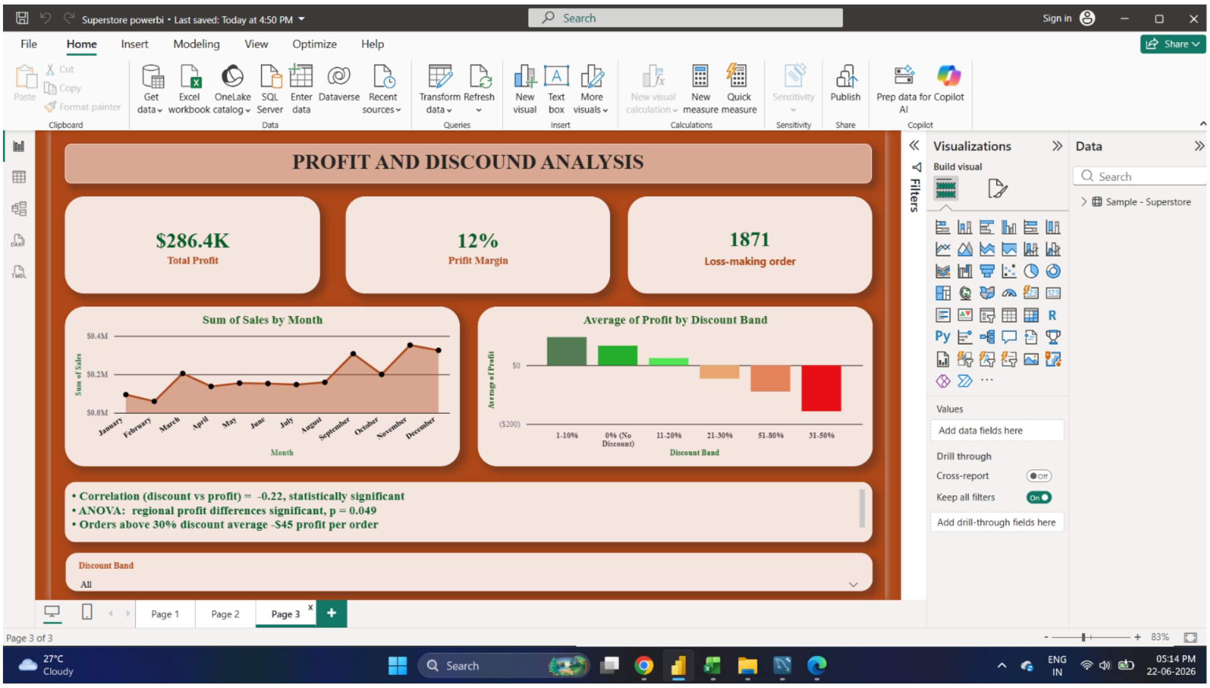


Figure 17: Product and Discount Analysis Dashboard

8 Key Findings

- Furniture generates revenue comparable to Technology ($742K vs $836K) but operates at a fraction of the profit margin (2.5% vs 17.4%).
- The Consumer segment generates the largest share of both revenue (51%) and order volume.
- Regional performance varies significantly: West outperforms Central by $16.76 in average profit per order, a difference confirmed as statistically significant by ANOVA ($p = 0.049$).
- Discounting above 30% is associated with negative average profit per order, while discounts of 1–10% are associated with the *highest* average profit — suggesting a non-linear relationship that pure visual inspection would miss.
- Monthly sales exhibit consistent Q4 seasonality across all four years, with November and December as peak months.
- Three sub-categories (Tables, Bookcases, Supplies) generate negative average profit despite meaningful sales volume.

9 Business Recommendations

Area	Recommendation
Pricing	Cap discounts at 20% for Furniture sub-categories; discounts above 30% should require managerial approval given their consistent negative profit impact.
Products	Reassess pricing or sourcing costs for Tables and Bookcases, which generate negative average profit despite strong sales volume.
Customers	Prioritise retention and upsell strategies for the Consumer segment, while exploring why Corporate spend concentrates in lower-margin Office Supplies.
Regions	Investigate and replicate the operational or pricing practices driving the West region's outperformance in the underperforming Central region.
Logistics	Review shipping mode allocation to reduce average days to ship without materially increasing cost.
Seasonality	Plan inventory and staffing ahead of the consistent Q4 demand spike observed across all four years.

Table 2: Business Recommendations

10 Conclusion

This project demonstrates a complete retail analytics workflow spanning data cleaning, feature engineering, statistical hypothesis testing, SQL querying, and interactive dashboard development. By combining Excel, MySQL, and Power BI with formal statistical validation — rather than relying on visualisation alone — the analysis converts 9,994 raw transaction records into a small set of statistically grounded, actionable business recommendations covering pricing, product strategy, customer focus, regional operations, and logistics.